Are Your Explanations Reliable?

Investigating the Stability of LIME in Text Classifiers

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Outline

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3. Preliminary Analysis
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Motivation

Modern ML systems are used in:

- healthcare diagnosis
- financial decision systems
- automated content moderation

However these systems are often:

Black-box models

Explainable AI (XAI) aims to make model predictions interpretable.

Example: LIME on Text Classification

Input Review

"The movie was visually stunning but the story was boring and predictable."

Black-box model prediction

$$f(x) = P(\text{Negative}) = 0.81$$

Step 1: Interpretable Representation

Words in the document:

$$(\text{movie}, \text{visually}, \text{stunning}, \text{story}, \text{boring}, \text{predictable})$$

Binary interpretable vector:

$$z' = (1, 1, 1, 1, 1, 1)$$

Interpretation

- Each feature corresponds to the presence/absence of a word.

Generating Perturbations

LIME generates nearby samples by removing words.

Perturbed Samples and Model Predictions

Perturbed text	Binary vector z'	$f(z)$
original text	(1,1,1,1,1,1)	0.81
remove "boring"	(1,1,1,1,0,1)	0.35
remove "predictable"	(1,1,1,1,1,0)	0.48
remove "stunning"	(1,1,0,1,1,1)	0.88
remove "boring,predictable"	(1,1,1,1,0,0)	0.22

Each perturbed sample is passed through the black-box model to obtain predictions $f(z)$.

Local Surrogate Model and Explanation

LIME fits a sparse linear model locally:

$$g(z') = w_0 + \sum_i w_i z_i$$

Estimated coefficients:

$$g(z') = -0.72 + 0.64(\text{boring}) + 0.48(\text{predictable}) - 0.30(\text{stunning})$$

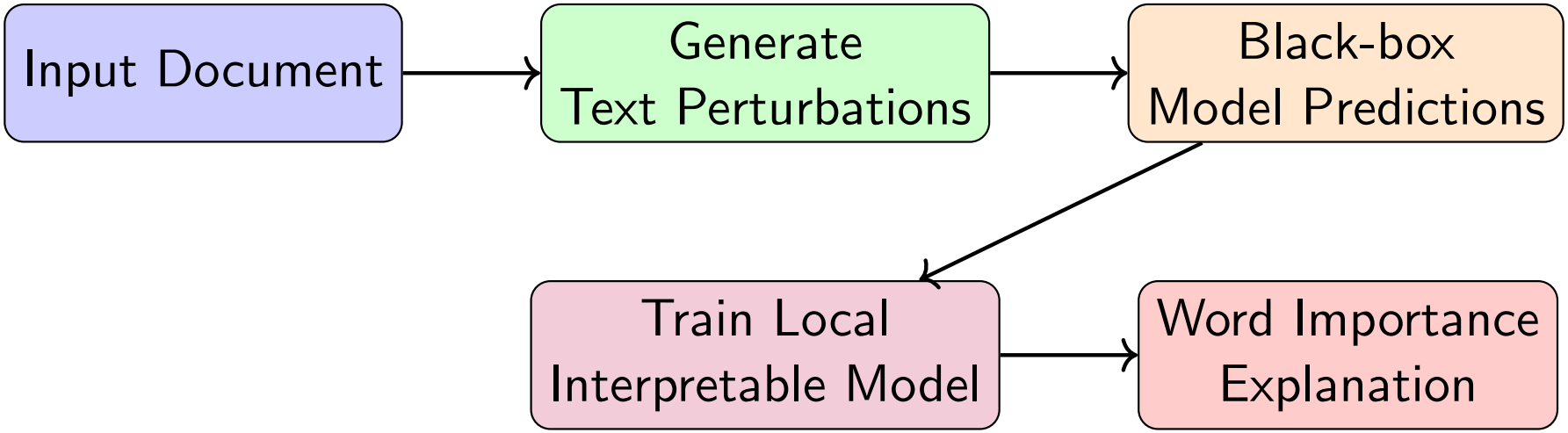
Explanation

- Strong negative contributors: **boring**, **predictable**
- Positive contributor: **stunning**

$\Rightarrow$ The negative prediction is mainly driven by

the words "boring" and "predictable"

LIME Explanation Pipeline



Key Property: Stability

A trustworthy explanation method should satisfy:

Stability

Small changes in input should produce small changes in explanations.

Otherwise:

- explanations cannot be trusted
- humans may draw incorrect conclusions

So, the question arises:

Are LIME explanations stable for text models?

Aim of the Paper

Goal

Study the **stability of explanations** produced by **LIME** for text classification models.

Baseline Analysis

- Examine the inherent stability of LIME explanations.
- Vary the **number of perturbation samples** used to train the surrogate model.
- Observe how changes in sampling affect the resulting explanations.

Proposed Method: XAIFOOLER

- Introduce **XAIFOOLER**, an algorithm designed to manipulate explanations.
- Iteratively perturbs a document through **word replacements**.
- Perturbations satisfy semantic and structural constraints.
- Goal: alter the explanation while keeping the classifier prediction largely unchanged.

Inherent Instability of LIME

Objective

Assess whether LIME produces **inconsistent explanations** even when the input and model remain unchanged.

Let

$$d = \text{document}, \qquad f(\cdot) = \text{target model}$$

We aim to explain the prediction: $f(d)$

Key Question

Does LIME generate different explanations for the same document even when the model and input remain identical?

This instability may arise due to the **random sampling process** used to generate perturbed documents for training the surrogate model.

LIME Explanation Generation Process

To explain the prediction $f(d)$, LIME performs the following steps:

Step 1: Generate Perturbations

- Randomly select k words from document d.
- Remove all occurrences of those words.
- This produces a perturbed document d_i.

Step 2: Sampling

Repeat Step 1 to generate

$$\{d_i\}_{i=1}^{n}$$

where n is the **sampling rate**.

Step 3: Train Surrogate Model

Train an interpretable model $g(\cdot)$ using

$$\text{Features: } d_i, \qquad \text{Labels: } f(d_i)$$

Impact of Sampling Rate

The parameter n determines the number of local samples used to train the surrogate model.

Small Sampling Rate (n)

- Insufficient perturbation samples
- Poor surrogate model $g(\cdot)$
- Unstable explanations

Large Sampling Rate (n)

- Better approximation of the target model
- More reliable explanations
- However increases computational cost

Observation

The randomness in sampling can lead to **different explanations for the same input**.

Evidence of Inherent Instability in LIME

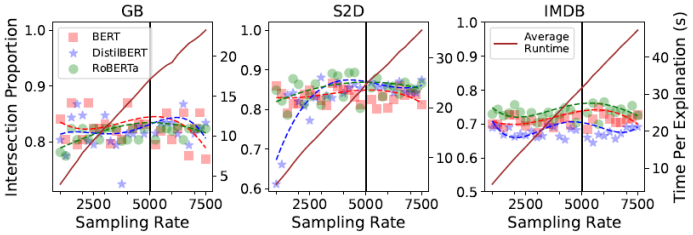


Figure 2: Inherent explanation instabilities for each model and dataset.

Dataset	Mean	Median	Min
GB	82.0% (↓ Δ18%)	82.0%	75.5%
S2D	84.0% (↓ Δ16%)	84.4%	72.9%
IMDB	71.5% (↓ Δ28.5%)	70.4%	67.3%

Table 1: Statistics of explanation similarities when using different alternate sampling rates compared against the base explanation with default sampling rate $n{=}5K$.

Key Observations

- Changing the sampling rate from the default causes significant drops in explanation similarity.
- Average similarity decreases by approximately 16% to 28.5%.
- Instability is more pronounced for **longer documents** because more training samples are required to approximate the target model $f(\cdot)$.
- Fig. 2 shows that explanation similarity stabilizes as the sampling rate increases, exhibiting **bounded variation and diminishing returns**.
- However, with **small sampling rates** (e.g. $n < 4K$ in IMDB), even slight changes in n can produce highly different explanations.

Adverserial Text Optimization Problem

Let

$$d_b = \text{base document}, \quad d_p = \text{perturbed document}$$

Let e_{d_b}, e_{d_p} = explanations produced by LIME
Objective: Generate a perturbed document that minimizes the similarity between explanations: $d_p = \arg\min_{d_p} \; Sim_e(e_{d_b}, e_{d_p})$

Interpretation

- Modify the document so that the explanation changes.
- The goal is to produce **very different explanations**.

Perturbations are performed through **iterative word replacements**, as commonly used in adversarial text attacks.

Constraints on the Perturbations

Unlike traditional adversarial attacks, we do not aim to change the model prediction.

$$\textbf{Prediction Constraint: } f(d_b) = f(d_p)$$

The perturbed document must produce the same prediction.

$$\textbf{Semantic Similarity Constraint: } Sim_s(d_b, d_p) \geq \delta$$

where

- $Sim_s(\cdot)$ measures semantic similarity between documents
- δ is a threshold ensuring meaning is preserved

Purpose

Ensure that the perturbed document remains **semantically similar to the original document**.

Limiting the Number of Perturbations

Even with semantic constraints, repeated perturbations may gradually alter the document meaning.

To control this, the number of perturbations is bounded:

$$i \leq \epsilon \cdot |f|$$

where

- i = number of performed perturbations
- $|f|$ = total number of unigram bag-of-words features
- ϵ = hyperparameter controlling the perturbation budget

Feature Overlap Constraint

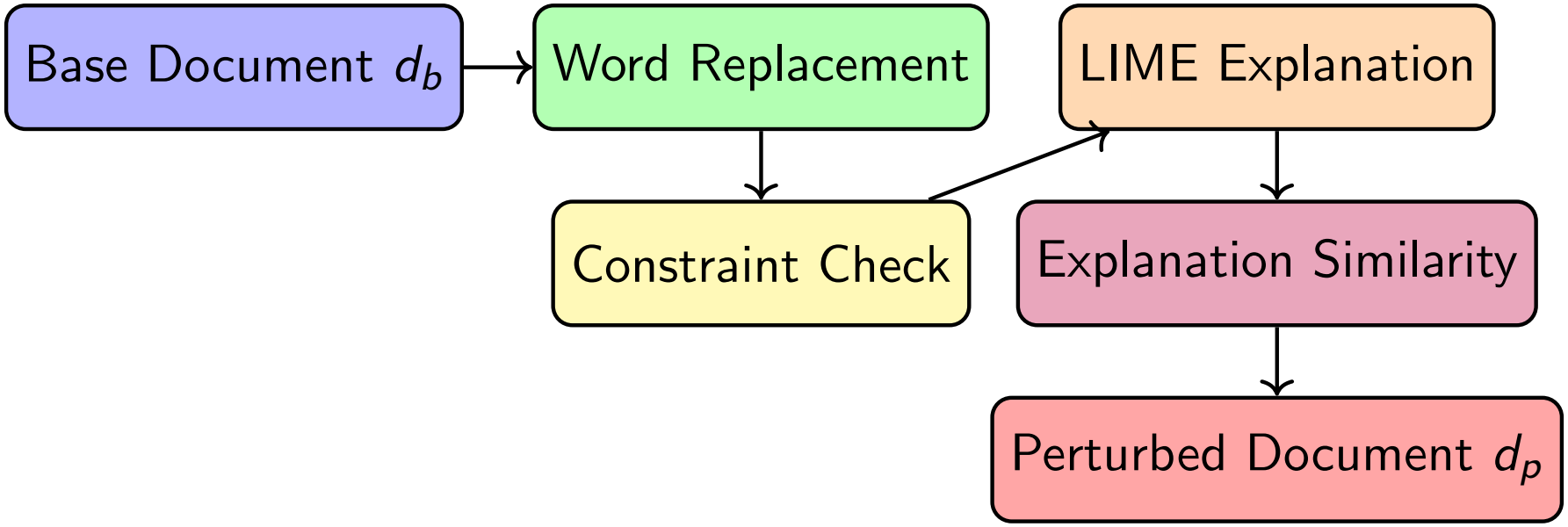
To preserve some relationship between the original and perturbed explanations, we require that at least one of the top-k features from the base explanation appears in the perturbed explanation.

$$e_{d_p} \cap c \neq \emptyset \quad \forall\, c \in e_{d_b}^{[:k]}$$

Interpretation

- $e_{d_b}^{[:k]}$: set of the top-k important features in the base explanation
- e_{d_p} : explanation generated for the perturbed document
- The condition ensures that the perturbed explanation shares at least one important feature with the original explanation.

XAIFOOLER Attack Pipeline



Objective function: $\arg\min Sim_e(e_{d_b}, e_{d_p})$

subject to

$$f(d_b) = f(d_p), \quad Sim_s(d_b, d_p) \geq \delta, \quad i \leq \epsilon \cdot |f|, \quad e_{d_p} \cap c \neq \emptyset \quad \forall\, c \in e_{d_b}^{[:k]}$$

XAIFOOLER: Overview

To solve the objective function defined earlier, the paper proposes **XAIFOOLER**, a greedy algorithm designed to manipulate explanations produced by text-based XAI methods.

Key Idea

- Explanations are represented as a ranked list of words ordered by their importance.
- The algorithm perturbs the input text so that the explanation changes significantly.
- The model prediction and semantic meaning of the document are preserved.

XAIFOOLER Algorithm: Step 1

Greedy Word Selection and Ordering

- Remove:
  - stopwords
  - top-k important features from LIME explanations
- Consider the remaining words in the document d_b.
- For each word, remove it individually and observe the change in the model prediction.
- Order words in descending order based on how much their removal changes the prediction.

Intuition

- Modify words with **lower predictive importance**.
- This highlights the instability of LIME explanations.

XAIFOOLER Algorithm: Step 2

Greedy Search with Constraints

- Replace each selected word with candidate substitutions.
- Accept a replacement only if:
  - explanation similarity decreases
  - prediction remains unchanged
  - semantic similarity constraint is satisfied
  - perturbation budget is not exceeded

Reducing the Search Space Candidate replacements must:

- maintain the same part-of-speech
- remain within a similarity threshold in Paragram-SL999 embedding space.(The Paraphrase Database (PPDB; Ganitkevitch et al., 2013) is an extensive semantic resource, consisting of a list of phrase pairs with (heuristic) confidence estimates.)

Note Although word-level perturbations may slightly affect text quality, XAIFOOLER is compatible with any black-box text perturbation function.

XAIFOOLER Algorithm

Algorithm 1 Adversarial Explanation Generation

Input: **target model** f, **Original Document** d_o, **Base Explanation** e_b, **Maximum Perturbation Threshold** p_t, **Current Perturbations** p_c **Current Similarity** s
Output: **Perturbed Document** d_p, **Perturbed Explanation** e_p, **Updated Similarity** s
Initialize: $e_p \leftarrow e_b, i \leftarrow 0, s \leftarrow 1, p_c \leftarrow 0$
Initialize: I $\leftarrow$ indices of replacement candidates that satisfy C
while $I \neq \varnothing$ *and* $p_c < p_t$ **do**
    s_i = RBO(d_p, perturb($d_p[I[i]]$)
    if $s_i < s$ **then**
        $s \leftarrow s_i$
        $e_p \leftarrow$perturb($d_p[I[i]]$)
    end if
    $I[i] = \varnothing$
    $i \leftarrow i + 1$
end while
return (d_p, e_p, s)

Why Not Traditional Similarity Measures?

Common measures fail:

- Jaccard: compares two sets by computing the ratio of the shared elements between the sets to the union of both sets.
- Kendall's τ & Spearman's ρ : ignores weights,the capacity to handle unequal list lengths, and disjoint features.

Need a similarity measure for ranked lists.
Solution:

Rank Biased Overlap (RBO)

RBO Mathematical Definition

RBO is defined as:

$$RBO = (1 - p) \sum_{d=1}^{\infty} p^{d-1} A_d$$

where

$$A_d = \frac{|L_1^d \cap L_2^d|}{d}$$

Interpretation:

- weighted similarity across ranking depth
- top features contribute more

Effect of Parameter p

$$p \to 0$$

focus on top-ranked features.

$$p \to 1$$

consider deeper ranks.(For depth D, $E[D] = \frac{1}{1-p}$)
Allows tuning importance of top explanation features.

Results

XAIFOOLER significantly manipulates explanations.
Findings:

- larger ranking shifts than baselines
- predictions remain unchanged
- semantic similarity preserved

Implication:
LIME explanations are vulnerable to adversarial manipulation.

Limitations

Several limitations remain:

- Only evaluates LIME (not SHAP or other XAI)
- Greedy algorithm may not find optimal perturbations
- Computational cost is high
- Dependence on surrogate model quality

Future Research Directions

Possible extensions:

- source of instability within LIME
- stability analysis of SHAP explanations
- theoretical bounds on explanation robustness
- defense mechanisms against explanation attacks
- robust explanation frameworks

Thank You